Second part: Open book

Total score: 10 pts.

You are warmly invited to be tidy and concise. Messy papers will be penalized.

5. A family has to plan the purchase of mineral water and wine for the next m pandemics days. The family knows the consumption of bottles of water (d_t) and of wine (d'_t) in every day $t = 1, \ldots, m$. W and W' are the number of bottles stored in the basement now (day 0). The wine and the water are delivered by two different suppliers. The delivery cost of water is c and the delivery cost of wine is c'. These costs are independent from the quantity and from the day. The overall capacity of the basement is u bottles (wine plus water). The problem of the family is to determine when and how many bottles of water and wine to purchase so as to minimize the overall delivery cost. Formulate the problem with a linear model. Specify the sets, the parameters, the variables, the constraints and the objective function, explaining their meaning.

   Variant: Let us suppose that the water supplier does not ask for the delivery fee if the purchased quantity is at least q bottles, and the supplier of wine does the same if the purchased quantity is at least q' bottles. How does the formulation change?

6. Consider an undirected complete bipartite graph $G = (N, A)$, where the set of nodes N is partitioned into tow disjoint sets S and T. That is $N = S \cup T$, $S \cap T = \emptyset$. S represents the set of nodes in the left column, and T those in the right column. The set of arcs is the Cartesian product $A = S \times T$. Each arc $(i, j) \in A$ has a cost $c_{ij} \geq 0$. A perfect matching M is a subset of arcs such that there is exactly one arc incident in each node. The minimum cost matching problem consist in finding a perfect matching whose sum of the costs is minimum. Provide a greedy algorithm for the problem. Thus describe E, F, and the functions $Best$ and Ind. Determine the arc costs in the small example represented in the picture for which your greedy algorithm does not produce the optimal solution.